Heaven’s Light is Our Guide

Rajshahi University of Engineering & Technology

Department of Electrical & Electronic Engineering

ASSIGNMENT

Marks		**Course No.**	EEE 4261
Allocated	**Obtained**	**Course Title**	Biomedical Signals, Instrumentation and Measurements
10		**Course Instructor(s)**	Md. Nuhi-Alamin, Md. Ariful Islam
		Due Date	17-06-2026
		Submission Date	17-06-2026
Student ID	2001019	**Student Name**	Mir Masrur

Problem Statement:

A patient experiences sudden cardiac arrest while on the way to the hospital. In this situation, the lack of real-time patient data made it difficult for the medical team to prepare in advance. Your task is to analyze the requirements for a biotelemetry system that can transmit critical patient data (e.g., ECG, heart rate, blood oxygen, blood pressure) from the ambulance to the hospital before arrival. This will help ensure the medical team is ready to provide immediate care. Your analysis should include the following:

- Select relevant biomedical sensors that can measure key physiological signals such as ECG, blood oxygen levels, heart rate, and blood pressure. Justify the choice of sensors based on their characteristics (sensitivity, accuracy, range, etc.) in relation to the specific needs of cardiac arrest management.
- Discuss the communication technology to be used for wirelessly transmitting the measured data, ensuring that it is reliable, fast, and can handle medical data securely.
- Define the components involved in the system (e.g., sensors, microcontroller, wireless transmitter, power sources, etc.) and their specifications.
- Explain how interdependencies between different system components (sensors, communication, and processing units) will be handled, with a focus on the system's robustness, reliability, and real-time performance.

General Instructions:

The solution of the assignment must contain:

1. ***Problem Statement:*** Provide a concise description of the problem to be addressed.
2. ***Proposed Biotelemetry System Model:*** Provide a detailed block diagram of the biotelemetry system, showing the data flow from the patient’s sensors to the ambulance communication system and eventually to the doctor in the hospital. Also, briefly describe each block of the proposed system.
3. ***Required Components:*** Identify and name all the components involved in the system (e.g., sensors, microcontroller, wireless transmitter, power sources). Provide detailed specifications for each component.

4. ***Cost Analysis and Feasibility:*** Provide a cost analysis for the system, considering the cost of sensors, communication modules, processing units, and other components. Discuss the economic feasibility of implementing such a system in real-world ambulance services and hospitals.
5. ***Risk Assessment and Mitigation***: Identify potential risks or challenges associated with the system, such as data loss, signal interference, or sensor failure. Propose mitigation strategies to address these risks and ensure system reliability in critical situations.
6. ***Ethical and Regulatory Considerations:*** Discuss the ethical and regulatory requirements for implementing such a biotelemetry system in a medical setting.
7. ***Addressing Complex Engineering Problems:*** Justify how your model satisfies P1 (Depth of knowledge), P3 (Depth of analysis required) and P7 (Interdependence).
8. ***Conclusion:*** Write the concluding sentences of this assignment task.
9. ***References:*** Cite the sources (journals/conferences/websites) used in IEEE referencing style.

Note: Points will be deducted from any submitted assignment that contains more than **30% plagiarism or AI-generated content**.

Addressing CO and PO:

Attribute	**Justification**
CO2: Analyze various biomedical sensor characteristics to select relevant sensor for a particular application and different biomedical signals. **PO(b):** Problem Analysis	Analyzing and selecting appropriate biomedical sensors based on their characteristics, such as accuracy, power consumption, and sensitivity, to ensure effective monitoring and real-time data transmission for wireless biotelemetry systems.

Addressing Complex Engineering Problems (P's):

Attribute	**Justification**
P1: Depth of Knowledge required	The problem requires in-depth knowledge of biomedical engineering, including an understanding of various biomedical sensor technologies, wireless communication systems and data handling for medical applications.
P3: Depth of analysis required	This task requires to use abstract thinking and originality to design a wireless biotelemetry system for critical biological data transmission management. There is no obvious solution, as it involves selecting appropriate sensors, communication technologies and ensuring system reliability under real-world constraints.
P7: Interdependence	The system's components must work together seamlessly, with each part (sensor, communication, and processing unit) interdependent on others to ensure real-time data accuracy and transmission.

Rubrics:

Source	Allocated Marks	Distribution of Marks			Obtained Marks
Proposed Biotelemetry System Model	4	**4:** Model is well-structured, high-resolution, with clear labeling with proper description.	**3-2:** Model is complete with minor issues in clarity, resolution, or labeling, or description is not adequate.	**1-0:** Model is incomplete, or very poorly written explanation.	
Required Components and Cost Analysis and Feasibility	3	**3:** All required components are identified and justified with a detailed cost analysis.	**2:** Some components' specifications or cost analysis lack detailing or missing	**1-0:** Incomplete components identification or limited cost analysis.	
Risk Assessment, Mitigation & Ethical and Regulatory Considerations	2	**2:** Clear identification of risks with detailed mitigation strategies. Ethical and regulatory considerations addressed thoroughly.	**1:** Some risks identified with basic mitigation strategies. Ethical and regulatory considerations discussed but lack depth.	**0:** Missing or minimal identification of risks or ethical considerations.	
Conclusion	1	**1:** Concise, and meaningful conclusion that reflects outcomes.	**0:** Very generic conclusion with minimal relevance or no conclusion.		
				Total Obtained Marks	

Problem Statement

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Proposed Biotelemetry System Model

The proposed biotelemetry model is a bridge between three important layer, patient body, the ambulance, and the hospital. Some sensors will be connected to the patient's body, through which biological data will be collected, and further processing and sending data will be done in ambulance and finally user end data will be received by the hospital for further preparation depending on the patient's condition.

ECG Sensor (AD8232): It will be attached to the patient body, this sensor will take heart electrical activity continuously, which is a small analog ECG waveform. This sensor will also convert it to heart rate.

SpO_2 Sensor (MAX30102): It’s a pulse oximeter, it will be clipped with the finger. Using light, it will measure blood oxygen level and send the data to the micro controller.

BP Sensor (Omron HEM-7120): It’s a cuff based non invasive blood pressure device that continuously measure systole and diastolic pressure.

Signal Conditioning (amplifier, low-pass filter, 12 bit ADC): This block works on raw ECG signal. The amplifier boosts the weak signals and low pass filter cancel out high frequency noise and motion artifacts from the ambulance and lastly the 12 bit ADC samples the cleaned waveform into digital signal and send it to the micro controller.

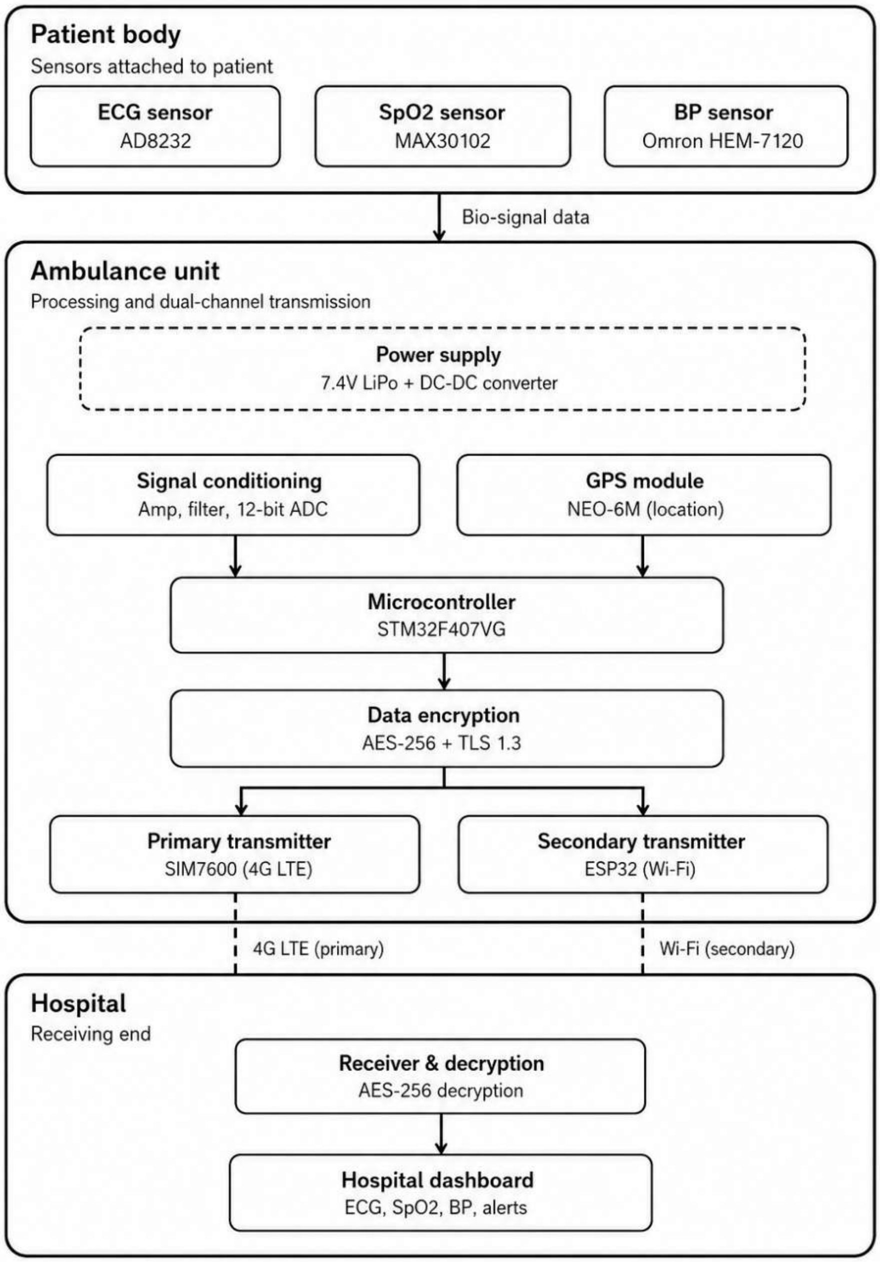

Figure 1: Block Diagram of a Wearable Biotelemetry System for Ambulance-to-Hospital Patient Monitoring

GPS Module (NEO-6M): This module determines the ambulance coordinates. Its output is send to the micro controller with the patients data so that the hospital dashboard can see the both condition. The ambulance live position and patients conditions.

Micro Controller (STM32F407VG): The processing core of the ambulance unit. It receives the digital data from signal conditioning and the location from the GPS Module and passes everything to the encryption block.

Data Encryption (AES-256 & TLS 1.3): Before any data leaves from the ambulance this block encrypt the data.

Power Supply (7.4V LiPo & DC-DC converter): This is the electrical foundation for every other block in the ambulance unit. The LiPo battery supplies 7.4V and the converter step that down to the 3.3V/5V for sensors, microcontroller, and the transmitters.

Primary Transmitter (SIM7600, 4G LTE): When the ambulance is away from the hospital, the ambulance communicate with the hospital through this unit. It uses LTE Cat-4 cellular network, which is fast and reliable.

Secondary Transmitter (ESP32, Wi-Fi): Once the ambulance near to the hospital it can transfer data using hospital wifi using this module.

Receiver and Decryption: At the hospital end this block accepts the incoming data and reverse the AES-256 encryption to recover the originals sensor and locations values.

Hospital Dashboard: This is the final block, that shows decrypted data as web based display for clinical staffs - with colour code alerts flagging abnormal readings, giving the medical team visibility into the patients conditions before the ambulance arrives.

Required Components and Specifications

Here we are attaching the requirement components and their key specifications. Mainly, these apparatuses are divided into these sections –

1. Sensors 2. Microcontroller 3. Medium/Channel 4. Modules

Components List:

Component	Model	Key Specifications	Usage
ECG Sensor	AD8232	CMRR > 80 dB, BW: 0.5-40 Hz, Supply: 3.3 V	Heart activity
SpO2 Sensor	MAX30102	Accuracy: ±2%, Resolution: 0.01%, I2C interface	Blood oxygen level
NIBP Module	Omron HEM-7120	Range: 20-280 mmHg, Accuracy: ±3 mmHg	Blood pressure
Heart Rate	AD8232	Range: 20-300 BPM, Accuracy: ±1 BPM	Cardiac rate monitoring
Microcontroller	STM32F407VG	168 MHz ARM Cortex-M4, 1 MB Flash, 192 KB RAM, 12-bit ADC	Data received and being processed
Cellular Module	SIM7600	LTE Cat-4, DL: 150 Mbps, UL: 50 Mbps, AT command interface	Wireless data transmission
Wi-Fi/BT Module	ESP32	802.11 b/g/n, BT 4.2, Dual-core 240 MHz	Short-range data transmission

Power Supply	LiPo Battery DC-DC	7.4 V/5000 mAh, step-down to 3.3 V/5 V	Portable power source
Encryption Module	Software AES-256	256-bit key, HIPAA-compliant TLS 1.3 tunnel	Data privacy
GPS Module	NEO-6M	Accuracy: 2.5 m CEP, Update rate: 5 Hz	Live ambulance location tracking
Display Unit	3.5 TFT LCD	320x480 px, SPI interface, sunlight readable	Dashboard

Cost Analysis and Feasibility

An estimated cost analysis is presented below. Here we show for 1 ambulance unit. The cost of these components was taken from available information online.

Component	Approx. Unit Cost (BDT)
AD8232 ECG Sensor	1315
MAX30102 SpO2 Sensor	345
NIBP Module	4450
STM32F407VG MCU	2250
SIM7600 4G LTE Module	5320
ESP32 Wi-Fi Module	427
NEO-6M GPS Module	430
LiPo Battery and DC-DC converter	530
3.5" TFT LCD Display	1640
PCB, enclosure, cables, misc.	3000
Software/development (one-time)	20000
Total Estimated Cost	~40000

Here, one noticeable fact is that when we are building a total environment, it needs BDT 40k for an ambulance, but our BDT 20k is used for software development, domain, and hosting purposes. It’s a one-time cost, so to install this system into another ambulance, we only need BDT 20k per unit. Which is feasible, as we are dealing with patients where a lack of treatment can cost a life. An ambulance can cost approx. 40 to 50 lakhs per unit, so adding some extra features to it with BDT 20k is normal and essential. Moreover, a commercial telemetry system could cost 55k BDT to 15 lakhs BDT, which is very expensive. Like Philips IntelliVue MX450 costs about 720000 BDT in the Bangladesh market [1].

According to the World Health Federation, approximately 377,200 deaths were recorded in 2021 for cardiovascular disease [2]. A telemetry feature can reduce this number because it can alert hospitals, doctors, and nurses to take precautions earlier.

Risk Assessment and Mitigation

Every projects/system is not fully independent. There are several risk factors that should be considered. Here, some issues are taken care of below.

1. **Sensor failure:** We are using various types of sensors here, like a heart monitoring sensor, a blood pressure sensor, etc. Sometimes these could malfunction and send wrong data to the hospital.
   Mitigation: Regular sensor calibration and a sensor alarm system for abnormal readings

2. **Motion noise:** Ambulance motion could affect ECG signals.
   Mitigation: We can use digital noise cancellation or adaptive noise cancellation.

3. **Network Issues:** The signal can be lost due to ambulance movement and rural areas.
   Mitigation: We are using a dual transmission system here; moreover, we could use an automatic retransmission system here.

4. **Incorrect Sensor Placement:** Improper sensor calibration by staff.
   Mitigation: We can provide a training session for the staff.

5. **Hospital Dashboard Failure:** Dashboard software could fail, or screen issues can occur.
   Mitigation: We can use a cloud-based system.

6. **Power Failure**: Telemetry is a portable system that works on DC power, so it could lose power at any time.
   Mitigation: We can use a DC-DC converter here that can use vehicle power.

Ethical and Regulatory Considerations

1. **Informed Consent:** If anyone is a cardiac patient and the condition is very critical, they may be unable to give consent on using his/her bio information. It is illegal to share patient information without consent. So, next of kin must be informed as soon as possible.

2. **Data Privacy:** Under the Digital Security Act 2008 in Bangladesh, all patient physiological data should be protected. That's why we propose an encryption model.

3. **Medical Device Regulation:** Any biotelemetry system must comply with medical device regulation. In Bangladesh, the DGDA, Director of Drug Administration, oversees these regulations. Our propose model use AD8232 and MAX30102 these are already certified for use..

Addressing Complex Engineering Problems

P1: Depth of Knowledge Required

The design of this biotelemetry system needs basic knowledge; it needs an analytical approach across several areas. To extract clear physiological data from a patient, must understand biomedical signal processing. This includes using amplifiers and low-pass filters to prepare analog signals before sending them through a 12-bit ADC. Additionally, protecting this medical data requires strong networking skills to implement AES-256 encryption and set up a TLS 1.3 handshake.

P3: Depth of Analysis Required

This system does not have an obvious easy solution and requires abstract thinking to formulate suitable models for a highly dynamic environment. For example, transmitting important information from a moving car is rather risky because the network can be weak, or the signal can be lost. This includes the analytical design of the dual transmission architecture, where the SIM7600 4G LTE module is used for primary communication for long distances and the ESP32 for secondary Wi-Fi transmission while in short range. In addition, advanced digital or adaptive noise cancellation techniques are to be implemented to analyze and mitigate motion noise, which distorts the ECG signals during transit.

P7: Interdependence

The proposed system is a high level problem with many inter-related sub-problems and components. It has critical link between three levels: the patient's body, the mobile ambulance environment and the static hospital infrastructure. The success of this system relies on the perfect sync of the biomedical sensors (like the AD8232 and MAX30102), the central processing microcontroller (STM32F407VG), a dual-channel wireless transmission and a secure web-based hospital dashboard that decodes and displays real-time alerts. The failure of any of the interdependent layers create problem can disturbs the entire medical preparation process.

Conclusion

This project shows a simple and cheap way to send patient data from an ambulance to a hospital. By using sensors attached to the body, we can measure things like heart activity, blood oxygen, and blood pressure. The system locks this data for safety and sends it using 4G mobile networks and Wi-Fi.

This is very helpful because hospital staff can see the patient's condition on a digital screen before the ambulance arrives. This means doctors can be fully ready to give treatment as soon as possible. Also, this system is much cheaper than big commercial machines. It costs about 40,000 BDT to build the first one, and only 20,000 BDT to add to more ambulances. Most importantly, alerting hospitals early with this system can save the lives of patients having heart problems.

Acknowledgment: The diagram (Figure-1) was designed with the assistance of Claude Sonnet 4.6 Model AI based on conceptual idea developed by me.

Reference

[1] Techno Power, "Philips IntelliVue MX450 Modular Patient Monitor," Techno Power. https://technopower.com.bd/product/philips-intellivue-mx450-modular-patient-monitor-1cyve

[2] World Heart Federation, "Bangladesh," World Heart Observatory. https://world-heart-federation.org/world-heart-observatory/countries/bangladesh/